Increasing Resolution of Diet-Microbe Relationships at the Compound-Enzyme Level Using Networks

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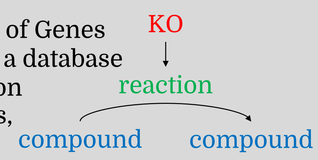
Introduction

Microbial studies produce data on both diet and microbiome composition and functional genes, but it is difficult to understand the relationships between them [1].

What previous research has found [1]:

- Western diet (high-fat, low-fiber) results in an increase in bacteria of the *Bacteroides* genus
- *Bacteroides* genus is associated with a decrease in short-chain fatty acids (SCFAs, beneficial) and an increase in Trimethylamine N-oxide (TMAO)
- TMAO is associated with an increased risk of cardiovascular disease

The Kyoto Encyclopedia of Genes and Genomes (KEGG) is a database containing information on genes (KOs), compounds, and reactions.



By leveraging experimental data (sequencing and dietary) and knowledge bases (KEGG) we are **developing methods to explore the relationship between microbes and the metabolism of dietary compounds**.

Compounds: include micronutrients, macronutrients, and small molecules (often referred to as non-nutrients).

Sample Collection and Processing

148 patients were recruited from:

- Two locations (rural and urban)
- Two timepoints 24 weeks apart

Three cohorts:

- ART naïve PLWH at week 0 (ART naïve)
- ART experienced PLWH (ART exp)
- Healthy controls (HC)

Surveys of diet in the form of 6-month recall Food Frequency Questionnaires (FFQ) were given at both times and 16S rRNA gene sequencing of fecal microbiome was performed. Initial analysis [2] showed *Bacteroides/Prevotella* differentiation which was correlated with patients in the rural location whose diet was more Agrarian (**Fig 1**).

Weighted UniFrac PCoA

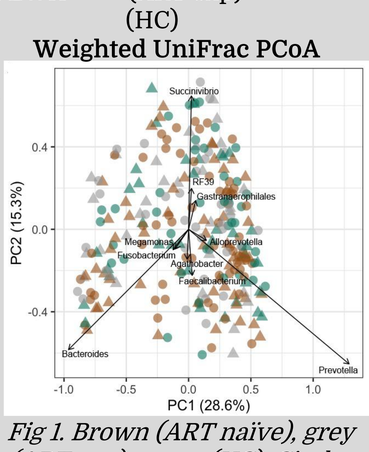


Fig 1. Brown (ART naïve), grey (ART exp), green (HC). Circles (week 0), triangles (week 24)

Food Type Biplot

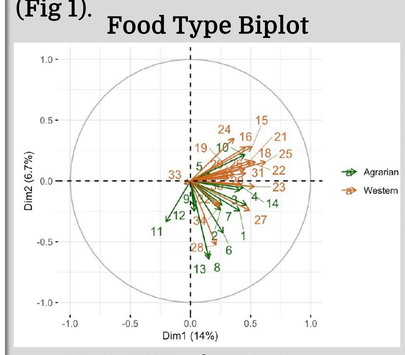


Fig 2. PCA of FFQ given to patients colored by diet

Of these 20 patients from the first visit were selected for shotgun metagenome sequencing, based on their diet (**Fig 2**).

Sequences were processed using HoMi, an open-source pipeline and "batteries-included" tool for streamlined data processing and analysis of dual-transcriptome and metagenome data.

HoMi Github

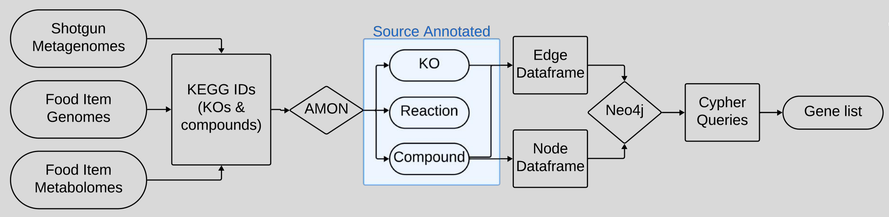
Objective

Using FFQ food items, shotgun sequencing from two patients, construct two graphs using KEGG representing Western and Agrarian diets and find instances of microbial metabolism of dietary compounds.

Problems

- From food item to KEGG compounds: **how do you define food?**
  - Use metabolomics databases such as *FooDB*
  - Use KEGG and get all KOs associated with a food item and *predict compounds* that might be present
- **How to determine reactions and compounds present from KOs and assign their origin?**
  - *AMON* [3]

Workflow



Graphs will be generated using Neo4j (a package developed for exploration and visualization of knowledge graphs) and explored using Cypher (query language)

Samples

ID	Cohort	Diet	WA Difference	Diet PCA2	Location	Prev/Bact Ratio	# of KOs
ZIM100.2	HC	Western	18	2.476	Rural	10.27	2,468
ZIM091.2	ART exp	Agrarian	-18	-3.187	Rural	2.48	3,597

FooDB

Food	Diet	Food Type
Milk (Cow)	Agrarian	Dairy
Butter	Western	Dairy
Cheese	Western	Dairy
Grape	Agrarian	Fruit
Orange	Agrarian	Fruit
Carrot	Agrarian	Vegetable
Sweet Potato	Agrarian	Vegetable
Chicken	Agrarian	Protein
Beef	Western	Protein
Eggs	Western	Protein
Corn	Agrarian	Grain
Bread	Western	Grain
Rice	Western	Grain

Resulted in 322 Agrarian and 227 Western associated compounds.

Whole Genomes

Western

Organism	# of KOs
cow	12,109
pork	11,791
bread wheat	4,102
rice	4,043

Agrarian

Organism	# of KOs
grape	4,138
carrot	4,053
chicken	10,407
corn	4,068

Experiments

#	Food Rep.
1	FooDB + Genomes
2	Genomes
3	FooDB

Agrarian Network Results

#	Microbe Nodes	Diet Nodes	Total Nodes	# of Patterns
1	289	1,120	2,174	426
2	296	1,120	2,174	444
3	945	109	1,054	826

KEGG Mapper results

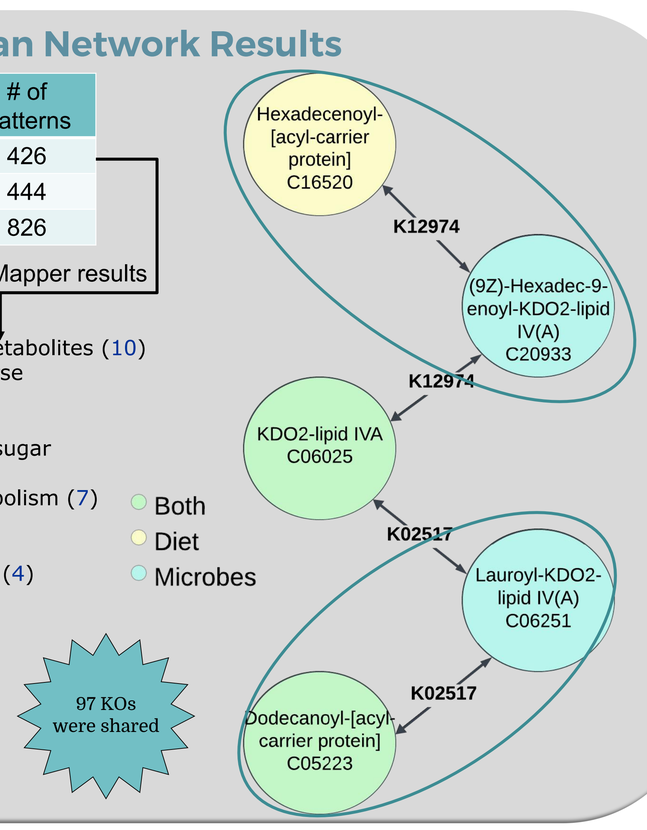
Global and overview maps
- 01100 Metabolic pathways (81)
- 01110 Biosynthesis of secondary metabolites (10)
- 01120 Microbial metabolism in diverse environments (15)

Carbohydrate metabolism
- 00520 Amino sugar and nucleotide sugar metabolism (11)
- 00051 Fructose and mannose metabolism (7)
- 00650 Butanoate metabolism (5)
- 00640 Propanoate metabolism (7)
- 00010 Glycolysis / Gluconeogenesis (4)

Energy metabolism
- 00910 Nitrogen metabolism (4)

Amino acid metabolism
- 00300 **Lysine*** biosynthesis (4)
- 00310 **Lysine*** degradation (3)

Metabolism of cofactors and vitamins
- 00790 Folate biosynthesis (3)
- 00860 Porphyrin metabolism (10)



Western Network Results

Global and overview maps
- 01100 Metabolic pathways (49)
- 01110 Biosynthesis of secondary metabolites (16)
- 01120 Microbial metabolism in diverse environments (8)

Carbohydrate metabolism
- 00010 Glycolysis / Gluconeogenesis (4)
- 00520 Amino sugar and nucleotide sugar metabolism (6)
- 00640 Propanoate metabolism (6)

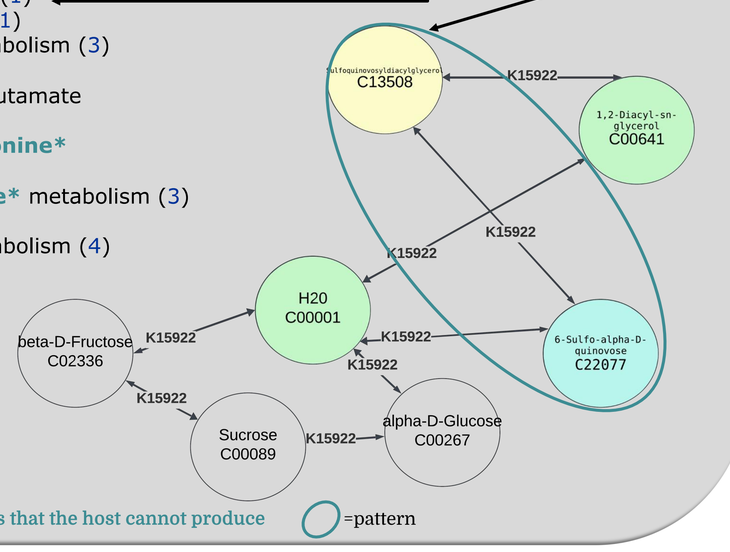
Lipid metabolism
- 00074 Mycolic acid biosynthesis (1)
- 00561 Glycerolipid metabolism (1)
- 00564 Glycerophospholipid metabolism (3)

Amino acid metabolism
- 00250 Alanine, aspartate and glutamate metabolism (2)
- 00260 Glycine, serine and **threonine*** metabolism (2)
- 00270 Cysteine and **methionine*** metabolism (3)
- 00310 **Lysine*** degradation (4)
- 00330 Arginine and proline metabolism (4)

#	Microbe Nodes	Diet Nodes	Total Nodes	# of Patterns
1	397	1,049	2,301	603
2	404	1,049	2,301	621
3	1,157	95	1,252	556

KEGG Mapper results

Sulfoquinovosyldiacylglycerol (SQDG)



*Essential amino acids: amino acids that the host cannot produce

=pattern

Discussion

Dietary observations:

- Metabolome → more small molecule resolution; higher likelihood of being digested
- Genome → limited to foods with a genome but there are more food items with genomes than with metabolomes
- Results show inputs do overlap in discovered patterns, but combination might do best

Comparing gene lists from identified patterns:

- Functional profiling of genes (KOs) from confirmed assumptions
  - Agrarian diets → metabolism of sugars
  - Western diets → metabolism of fats
  - Metabolism of essential amino acids was represented

A closer look at a Western associated KO:

- SQDG is typically found in the membranes of photosynthetic species (e.g., leafy green vegetables)
- Herbivores acquire it upon consumption
- Some *Proteobacteria* can catabolize it [4]
- Upstream of hydrogen sulfide → health benefits [4]

Despite the occurrence of this in the western diet could be a product of overlap in diets.

Computational Goals:

- ☑ Develop framework for graph representation of microbial metabolism of dietary compounds.
- ☑ Leverage Neo4j+Cypher for easy query structure for pattern search
- ☑ Understand limitations of current data bases

Biological Goals

- ☑ Experiment on different representations of diet and the resulting graphs
- ☑ Confirm known mechanisms of microbial metabolism
- ☑ See differences in metabolism between diet types confirming known assumptions

Future Directions

Enhancing the capabilities of AMON
- Currently it only confirms whether metabolites in metabolome sample are predicted and does not query KEGG for the others

Connecting KEGG → not all KOs have a reaction despite having an EC number

Further validation → MICOM
- See if MICOM predicts same compound exchange between species that is seen here

More investigation into differences in diet input type
- Identify lack of patterns when using genomes
- Identifying similarities
- Comparing plant versus animal consumption

Addressing some challenges
- Could the addition of other databases and ontologies result in a more informed graph?
- Addressing the removal non-informative compounds such as H2O

Building a graph of "all known" knowledge
- All FooDB + All microbe associated KOs → graph → all possible connections
- Could function as a positive control

Utilizing more samples to compare diets
- Can build a graph for each of the 20 patients
- Compare similarity of graphs between patients who are more Agrarian vs Western
- Perform PERMANOVA

Diet-Microbe-Net Github

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References

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